On the discussion questions regarding Frege's 'On Sense and Reference' and Kripke's 'Identity and Necessity' (December 24, 2025)

Here are a couple of thoughts and observations on the discussion questions. As always, comments are welcome!

All groups 1:
'Fregean sense determines both reference and cognitive content. Intuitively, it does not follow that knowing the sense of an expression implies knowing what it refers to. Discuss. What would be Kripke's stance?'

Fregean sense determines cognitive content at the sub-sentential level as well as at the level of sentences. The former is illustrated by solution of the problem of the cognitive status of identity-statements. The latter follows from compositionality: it is the sense of sub-sentential expressions that determines the sense of sentences, i.e., the thought that they express. At the same time, sense also determines reference: it is in virtue of its sense that a name (proper names and definite descriptions) refers to this object rather than that. And likewise, it is the sense expressed by a sentence that determines that it is true, rather than false.[1]

The fact that sense is stable and that reference can change is what makes cognitive content substantial. The sense of a name characterises what it refers to in a particular way and that enables us to pick out an object as the reference, by checking which unique object (if any) has the properties that the sense specifies. In different circumstances that may be a different object, picked out by the same sense. In an analogous way the thought expressed by a sentence enables us to determine its truth-value, by checking whether reality conforms to the thought, i.e., that reality is as the thought says it is. And the same thought may be true in some circumstances, false in others.

The point to note is that sense by itself is not enough to enable us to do these things, at least not in most cases. In addition to grasping the sense of an expression we also need to know enough about the circumstances to be able to identify a particular object as the referent of a name, or to determine a particular truth-value as the reference of a thought. We need the cognitive content of the expression *and* enough knowledge about the relevant aspects of the circumstances (and these many be actual or possible, present, past, or future) to be able to 'proceed from sense to reference'. So, sense is only part of the equation.

This seems to square with intuition. It seems obvious that knowing the meaning of an expression does not guarantee knowing its reference. We may understand what someone says without knowing whether what they say is true or not, or what the expressions they use to formulate their thought refer to.[2]

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[1] Frege's analysis of sense and reference of predicates and relation symbols can be found elsewhere, in 'Function and Concept' ('Funktion und Begriff', 1891) and in 'On Concept and Object' ('Über Begriff und Gegenstand', 1892). English translations by Peter Geach and Max Black can be found in Geach & Black, *Translations from the Philosophical Writing of Gottlob Frege*, 1960 (reprinted in Beaney (ed), *The Frege Reader*, Blackwell, 1997).

[2] This is not to say that this complete separation between linguistic knowledge (knowing what expressions mean) and world knowledge (knowing what expressions refer to) is obvious in each and every case. One could very well argue that understanding linguistic meaning in some cases implies knowing the referent or minimally knowing something more about the referent than is given by the linguistic meaning. Consider knowing what 'electron' means, or 'beech'. If I cannot tell a beech from an oak, a spruce, a sequoia, or any other kind of tree, can I say that I know what 'beech' means? In such cases linguistic meaning has a practical ('know how') component in addition to a cognitive ('know that') one. See Putnam's seminal paper 'The Meaning of "Meaning"' (1975) for arguments along these lines.

Intuitive as it may seem, rigid designation complicates this picture. If a name designates rigidly, it has a sense that picks out the same referent in all circumstances in which it refers. Such a sense is a constant function from possible circumstances to a fixed object. Does it follow that such a sense does not contribute to cognitive content? That depends. Proper names for people or cities or other types of concrete objects usually do not, since such proper names do not characterize their referents in a way that depends on descriptive content.[3] These rigid designators are indeed meaningless 'tags'. But there are also definite descriptions that refer rigidly, for example descriptions of mathematical objects such as 'the square root of 16', or 'the largest prime smaller than 10 to the power of 6'. Such an expression refers to the same object in every circumstance, so it is rigid. But it does so in virtue of the sense of its component expressions, i.e., it rigidly refers by *describing* its reference in terms of its properties. That means that this type of rigid designator does have cognitive content and, thus, contributes to the cognitive content of sentences in which it occurs.

Now what about the distinction between knowing the sense and knowing the reference that arises with Frege's analysis? With 'tag'-type rigid designators it appears that there is no distinction since there seems to be no cognitively relevant difference between a constant function to an object and the object itself: knowing one is indeed knowing the other. Thus, as far as determining the referent of 'tag'-type rigid designators is concerned it is indeed the case that either we know what they refer to or we don't. However, that does not necessarily percolate up to the sense – reference connection of complex expressions in which 'tag'-type rigid designators occur, such as sentences. Only sentences that consist entirely of this type of rigid designator plus some logical vocabulary (identity, quantifiers, connectives) exhibit a similar breakdown of the distinction between knowing the sense and knowing the reference. In all other cases, the occurrence of descriptive phrases, such as nouns, verbs, prepositions, adjectives, adverbs and the like, will turn the sense of a sentence into something that calls for additional, non-linguistic, knowledge of circumstances in order for us to be able to determine its truth value.

That holds for 'tags'. But what about the other type of rigid designator? Note that in the case of rigid designators that have cognitive content their rigidity is not due to the nature of the referent. Take 'the square root of 16'. It refers to a number, but whatever special ontological status we might want to give to numbers that by itself does not make 'the square root of 16' rigid. For consider 'the number I am thinking about now': that, too, refers to a number, but it is not a rigid designator. So, the rigidity of 'the square root of 16' is due to its descriptive content, just as it is the descriptive content of 'the number I am thinking about now' that makes it a non-rigid designator. The nature of the referent is irrelevant.

So, the relation between knowing the sense of a name and knowing its referent is much more complicated. It is not just a matter of the overall status of a name, i.e., its being rigid or non-rigid. What also matters is its component expressions. Ultimately that is what determines what a name contributes to the sense of the expressions in which it occurs.
Thus, the connection between knowing the sense of a sentence that contains rigid expressions and knowing its truth-value depends also on the nature of the rigid expressions: do they contribute sense or not? Note that compositionality is at play at every stage here.

[3] In some languages proper names have more descriptive content than in others, but even where they do, that content is not used to *identify* the bearer of the name. In languages such as English proper names often go back to Latin or Greek phrases that are meaningful *in Latin or Greek*. But that meaningfulness does not transfer to their use as proper names in English.

All groups 2:
'Sometimes cultural artefacts are repaired or restored using different materials than those from which the original was made. Evaluate this from the perspective of Kripke's idea of 'necessity of origin'. Is necessity of origin more plausible with regard to natural entities than with regard to artefacts?'

If we take Kripke's stance on the lectern made of ice as our starting point, it seems that any modification of the 'stuff' that a material object is made of changes its identity. But there are complications even in what looks like a straightforward case. It seems reasonable to hold that replacing some part, say the left-leg, of the ice-lectern by a leg made of wood, turns it into a different object. But what if we replace the left-leg by another piece of ice of the same shape and dimensions? Do we still end up with a different lectern? Even if the replacement leg is frozen from the same water that the formed the ice of the original leg?[4]

Nowadays restorers of cultural artefacts wrestle with these questions. But they have not always done so. In the past painting over parts of an old painting to enhance its colours, or to add a background figure, or, reversely, to hide parts of it (many nudes were painted over at some point in time) were considered perfectly acceptable ways of dealing with it and nobody would consider them as (drastically) changing the identity of the painting. Nowadays, those practices have been largely abandoned and today's restorers try to abide by the principle of 'reversibility and authenticity' as much as possible: any change brought about by restoring an artefact must be reversible, i.e., it must be possible to undo it in the future, and it must be done using materials that come as close as possible to the original ones.[5]

That is easier to do when dealing with such artefacts as paintings, jewellery, pottery, furniture. But with an increase of scale things becomes much more complex: it is nigh impossible to restore a building in a way that satisfies the reversibility and authenticity principle.[6]

At the background of these considerations are questions about identity. In many societies buildings used to be constructed from wood and wood is susceptible to fire. If a building burns down and it is rebuilt using the same kind of wood according to the same plan, in all details, using the same techniques, is it the same building? Not according to Kripke's strict criterion, it seems. And even if we lower our standards and drop the requirement that we cannot change the actual material, we still face a question. Suppose a building was erected in 1600, burned down in 1670, and then was rebuilt 1675 in the strict way described above (same plan, same techniques, same kind of wood): is it the same building? We say: 'The building burned down in 1670 and it

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[4] We can even do away with the lectern. Take an ice cube container containing an ice cube from the freezer compartment of your refrigerator. Let the ice melt. Put the container back in the freezer. What's the result: the same ice cube? A different one? The material composition is the same. Its properties along the object's spatiotemporal trajectory are not. So?

[5] 'Largely abandoned': for sometimes things do go wrong: https://www.independent.co.uk/arts-entertainment/art/news/elderly-woman-who-destroyed-19thcentury-spanish-fresco-says-i-didn-t-do-it-secretly-8076528.html.
We would do well to note that not only have our criteria here changed over time, there is also no reason to think that they will not continue to do so. There is indeed growing consensus globally with regard to a set of standards. But it would be presumptuous to assume that the standards we go by today are set in stone. (And note also, that for the general public these standards may very well lack the kind of normativity that they have for professionals and connoisseurs.)

[6] An interesting case is provided by purely digital artefacts. Here material composition is simply irrelevant, which means that any form of identification in terms of material composition is useless. So, what determines their identity? What modifications keep that identity intact, what change it? At this point we enter the mysterious world of the trade in non-fungible tokens …

was rebuilt in 1675.' The use of 'it' suggests that we refer to something over and above the actual buildings (old and new), to something that has remained the same. But notice that we cannot say of the building after 1675 that 'it has been here since 1600', for between 1670 and 1675 there wasn't anything there. That suggests that spatiotemporal continuity is a necessary criterion for something to exist as 'the same'.

But then other cases suggest themselves, one of the most famous being that of Theseus' ship. According to Greek legend king Theseus on a long journey replaced the parts of the ship he was sailing on piece-meal: one plank today, another next week, a bit of rope after that, and so on. (It was a very long journey …) Was the ship that Theseus arrived in the same as the one that he set sail in? This example has been debated for centuries (literally), and there seems to be no real agreement about what to say in a case like this.

And an even more telling example (if only because it does not involve Greek mythology …) is that of Ise Jingu, a Shinto shrine in Japan, that has been torn down and rebuilt every twenty years for more than 1300 years.[7] It that the same shrine still, after sixty-three times of tearing down and rebuilding? The architectural lay-out remains the same, the construction methods do not change, nor does the function of the buildings. But the materials are new, and the rebuilding takes nine years, so there isn't substantial spatio-temporal continuity.

However, in terms of its *function* it is in an important sense 'the same shrine', with a continuous history of more than 1300 years. So, one could ask, why not replace the concept of identity of an object *x* with that of identity of what serves the function of *x*? Will that do? The fact that the paraphrase is puzzling might indicate that the answer is not an unqualified "Yes". For what serves the function of *x*, i.e., of the original object? As long as the original object *defines* the function, we have not rid ourselves from it, and thus still face the question of its identity. Of course, we can loosen the connection and just say we need an answer to the question after the identity of 'whatever serves the functions that were served initially by the object'. Then the original object drops out of consideration.

However, its materiality still matters, the concept of identity does not become purely 'ideal'. These objects being material and having material properties, is one of their key characteristics. We are what we are also (not exclusively of course) because of our bodies; a painting is what it is because it can be seen; a building is what it is because we can look at it from the outside and from the inside; and so on. Without their materiality these objects lose a large part of their identity, of what they are for us.

That being said one can think of cases where a functional perspective is certainly a possible, perhaps even the most intuitive way of thinking about identity. Cases in point are entities such as institutions, corporations, states, … Here function seems radically separated from any form of material embodiment. But do note that spatio-temporal continuity may play a role here as well. For example, the physical location of a company's headquarters may determine its legal status. And is a nation state that comprises of substantially different territory still the same nation state?[8]

Now what about the identity of humans? We know that the cells of which the human body is composed change: old cells are constantly replaced by new ones. The pace at which this happens is not the same for every type of cell: fat and muscle cells change at the slowest pace, 12 and 50 years on average; but cells in our gut usually change at a pace of a few days to a week. In any

[7] https://www.independent.co.uk/asia/japan/japan-ise-jingu-shrine-visit-rebuilt-b2831812.html

[8] Poland is an interesting example in this respect; neither its geographical territory nor its temporal existence is really continuous.

case, once you are past a certain age there isn't much 'original material' left. Yet we would not say that a person changes into another person simply because of this. (We do of course talk a lot about changes that people go through and even say things like 'He has become a completely different person', but I assume that by that we do *not* mean that someone has stopped being and someone else has come into existence instead.)

That type of change in the body thus seems not very relevant. But what about brain cells? What if we identify, not with the entire body, but with the brain? C*ommunis opinio* has it that in adulthood no new brain cells are generated.[9] If this is the case, it raises the following question: if we identify ourselves with our brains, and our brains only degenerate (in the sense of cells dying and not being renewed), what does that say about us? That we change? That we diminish? Or that we need to re-think our identification with our brain?

Another approach to personal identity also proceeds at the biological level: we could go for the DNA-profile. This is indeed a very effective way of identifying people. But who of us 'self-identifies' with their DNA-profile? It seems not a very likely thing to do, if only because, unlike with our bodies and in a certain also with our brains, we are not familiar with our DNA-profile. We know we have one. And a few of us might even know what it looks like. But we lack the kind of intimate and inward knowledge that seems essential for our sense of identity.

And then there are people (usually, very rich ones ...) who claim that dumping the contents of their brains onto a machine is them surviving death. But would many of us concur? Notice that we are dealing here with two reductions at the same time: of ourselves to our brains, and of our brains to whatever parts of that survive the dump onto a physical data carrier. I would venture that this is a concept of identity that very few of us would want to apply to ourselves.

So, it seems that in everyday life we work with a different type of identity, which is often called 'narrative identity'. We identify with where we were born, how and where we grew up, what changes we went through in our youth and adolescence, and so on and so forth. That is a type of identity that is far removed from what we understand by it from a logical point of view. But it is arguably the most important concept of identity that we have.

What does and does not play a role in our narrative identity is not fixed. For example, it may depend on a person's history, their knowledge of that history, their knowledge of science, and so on. Nowadays many people would be inclined to think about themselves more in terms of biological parameters than, say, fifty years ago, when science hadn't progressed as much as it has today and, also important, the general public knew much less about these results. What is interesting, of course, is that for most people it isn't actual scientific knowledge, but very general, coarse-grained and often only half-understood 'facts' that go into how they think about these issues.

[9] Apparently, the jury is still out there on the question whether neurogenesis (generating of brain cells) occurs in adults. Some studies suggest that adult neurogenesis might occur in the hippocampus. Other studies claim that if it occurs at all, it occurs at undetectable levels.